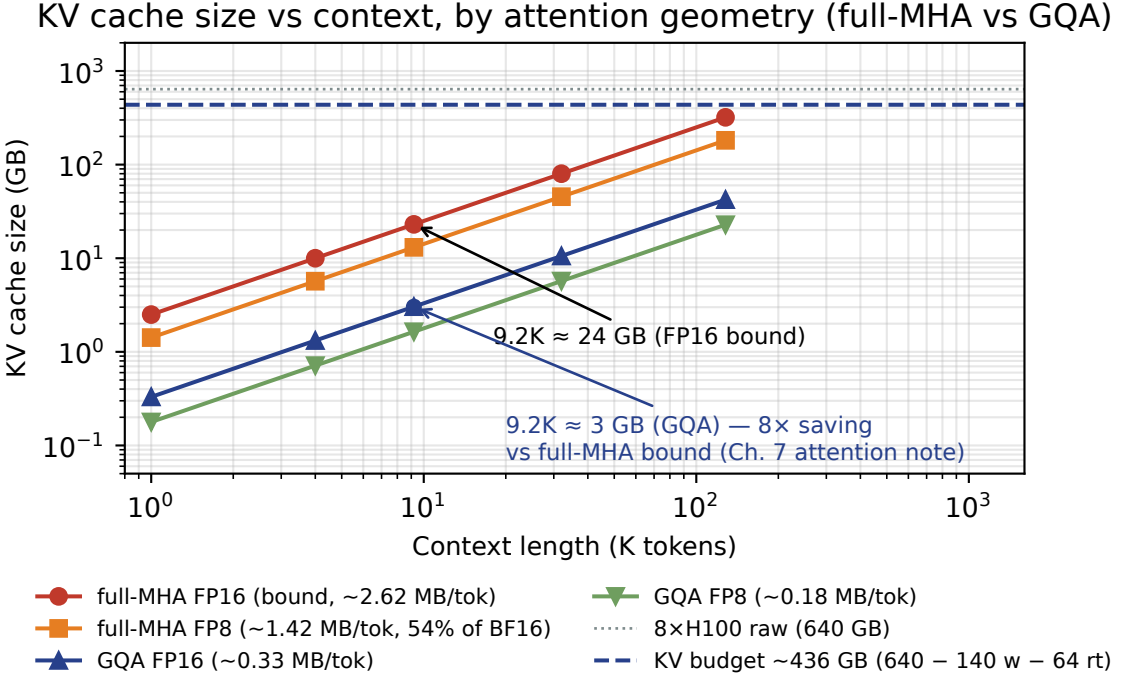
KV cache size vs context, by attention geometry (full-MHA vs GQA)
KV cache size (GB)
Context length (K tokens)
9.2K ≈ 24 GB (FP16 bound)
9.2K ≈ 3 GB (GQA) — 8× saving
vs full-MHA bound (Ch. 7 attention note)
full-MHA FP16 (bound, ~2.62 MB/tok)
full-MHA FP8 (~1.42 MB/tok, 54% of BF16)
GQA FP16 (~0.33 MB/tok)
GQA FP8 (~0.18 MB/tok)
8×H100 raw (640 GB)
KV budget ~436 GB (640 − 140 w − 64 rt)